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An evaluation of multi-species occupancy models with correlated species occurrences

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Abstract

1. Multi-species occupancy models for estimating the effects of environmental covariates on species occurrences while accounting for the effects of false-negative errors in detection were developed more than 20 years ago. Only recently have these models been extended to include correlations in occurrence between species.
2. Tobler et al. (2019) proposed two of these models wherein species occurrences are specified using a probit-regression model of correlated multivariate binary outcomes. In one model, correlations in occurrence between species are formal parameters. In the other model, a factor-analytic approximation of these correlations is used. After conducting extensive simulation studies to compare the performance of these models, Tobler et al. (2019) recommended the latter model in favour of the former owing to difficulties in fitting the former to simulated or real data. The same software (JAGS) was used to fit both models.
3. Since the former model specifies the actual process used to generate species occurrences, we hypothesized that shortcomings in the software were responsible for the difficulties reported by Tobler et al. (2019). We therefore devised an efficient Markov chain Monte Carlo algorithm for fitting the data-generating, multi-species occupancy model using parameter-expanded data augmentation. Using this algorithm, we conducted analyses of simulated data sets to re-evaluate and compare the performance of the two multi-species occupancy models. We also compared the results of fitting these models to a real dataset collected in camera-trap surveys of mammalian species on the Tibetan Plateau.
4. Our analyses of simulated data revealed that estimators of species occurrence, correlation and detection parameters had similar performance in terms of average bias, average root-mean-squared error and average coverage of 95% credible intervals regardless of whether the data-generating model or its factor-analytic approximation was used to analyse the data. However, our analyses also revealed the presence of systematic bias in the correlation estimates of both models, wherein strongly negative correlations were overestimated and strongly positive correlations were underestimated. Systematic bias was not evident in

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correlations estimated by fitting a multivariate probit regression model to latent species occurrences; therefore, the systematic bias in correlation estimates of the two occupancy models must have been produced by the failure of these models to estimate latent species occurrences accurately.

KEYWORDS
joint species distribution models, latent-variable models, multivariate probit models, occupancy models

1 | INTRODUCTION

Single-species occupancy models (MacKenzie et al., 2002; Tyre et al., 2003) are widely used in ecology to estimate and predict the spatial distribution of a species from data collected in presence–absence surveys. The popularity of these models stems from their ability to estimate the effects of environmental covariates on species occurrences while accounting for false-negative errors in detection, which are ubiquitous in surveys of natural populations. Building upon these models, multi-species occupancy models (Dorazio et al., 2006; Dorazio & Royle, 2005; Gelfand et al., 2005) were developed for the analysis of presence–absence data collected in surveys of communities of ecologically similar species (Dorazio et al., 2011). An important benefit of these models is that they can be used to estimate species richness and other measures of biodiversity because the models predict occurrences of species that were present but not detected in any of the surveys.

The development of models of presence–absence data that estimate species occurrences while ignoring the effects of detection errors followed a progression similar to that of occupancy models. Initially, various types of univariate, binary-regression models were adopted for use as species distribution models (SDMs) of a single species (Elith & Leathwick, 2009). Subsequently, multivariate binary-regression models were adopted or extended to analyse presence–absence data collected in surveys of multiple species (Latimer et al., 2009; Ovaskainen et al., 2010, 2016; Pollock et al., 2014; Thorson et al., 2015; Tikhonov et al., 2017; Warton et al., 2015). These models are referred to as joint species distribution models (JSDMs) to distinguish them from their single-species counterparts. JSDMs are designed to estimate the effects of environmental covariates on the spatial distribution of all species observed during sampling. Unlike the multi-species occupancy models that were developed earlier, JSDMs also provide estimates of correlations in occurrence between species after controlling for the effects of environmental covariates. Although the source of these correlations cannot be determined unambiguously (e.g. they may be attributed to interactions between species or to the effects of unobserved environmental covariates), correlations between species do provide measures of co-occurrence between species that cannot be attributed to the effects of observed covariates. So JSDMs are quite useful in this regard. In addition, some JSDMs include spatially correlated random effects that can account for unobserved sources of spatial variation in species occurrences (Ovaskainen et al., 2016; Thorson et al., 2015; Tikhonov et al., 2020).

Following the development of JSDMs, multi-species occupancy models were proposed to include correlations in occurrence between species while accounting for false-negative errors in detection (Doser et al., 2023; Hogg et al., 2021; Tobler et al., 2019). The occupancy model of Doser et al. (2023) adopts the approach taken in most JSDMs by modelling the probability of occurrence of each species as a function of fixed effects of measured environmental covariates and random effects designed to account for correlations between species and unobserved sources of spatial dependence. In contrast, the multi-species occupancy models proposed by Tobler et al. (2019) and Hogg et al. (2021) are formulated somewhat differently. In these cases, latent species occurrences are modelled using Chib and Greenberg's (1998) probit-regression model of correlated multivariate binary outcomes. By adopting this approach, the correlations in occurrence between species are induced without requiring overdispersion in the marginal distribution of each species' occurrence. The same approach was used in the JSDM of Pollock et al. (2014).

In the present article, we focus on an evaluation of two multi-species occupancy models proposed by Tobler et al. (2019) for estimating correlations in occurrence between species from data collected in presence–absence surveys of communities. Although both models are based on the multivariate probit-regression approach of Chib and Greenberg (1998), one model is parameterized in terms of correlations between pairs of species, whereas the other model is parameterized using a factor-analytic approximation of these correlations. Tobler et al. (2019) distinguished these models by the abbreviations MP (multivariate probit) and LV (latent variable), respectively. After conducting extensive simulation studies to compare the performance of these models, Tobler et al. (2019) recommended use of the LV model in favour of the MP model because they found difficulties in fitting the MP model in communities of more than 10 species. Although the same software program, JAGS (Plummer, 2025), was used to fit both models, Tobler et al. (2019) concluded that the Markov chain Monte Carlo (MCMC) algorithm created by JAGS to fit the MP model 'showed poor convergence, very slow mixing and was often unable to accurately estimate parameters' despite having used 250,000 MCMC iterations, discarding the first 200,000 iterations as 'burn-in', and thinning the remainder

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of the Markov chain by retaining every 50th iteration to estimate summaries of the posterior distribution. These difficulties are consistent with those of Pollock et al. (2014), who used JAGS to fit their JSDM model while using 1000,000 MCMC iterations, discarding the first 15,000 iterations and thinning the remainder by retaining every 1,000th iteration.

The recommendation of Tobler et al. (2019) to favour the LV model over the MP model is understandable given the results of their simulation studies. However, further evaluation of these multi-species occupancy models seems warranted before dismissing the MP model as too difficult to fit. After all, the MP model is based on the actual correlation matrix, not an approximation of this matrix as in the LV model. In the present study, we reformulated the MP model using parameter-expanded data augmentation (Liu & Wu, 1999) for the purpose of deriving an efficient MCMC algorithm to fit this model. We then conducted analyses of simulated data sets to re-evaluate and compare the performance of the two multi-species occupancy models proposed by Tobler et al. (2019). Our simulation-based evaluation was focused on estimating the correlations in occurrence between species and the parameters used to calculate species occurrence and detection probabilities from environmental covariates. We also compared the results of fitting the MP and LV models to a real dataset collected during a camera-trap survey of mammalian species on the Tibetan Plateau.

2 | MATERIALS AND METHODS

In Sections 2.1 and 2.2, we describe the multi-species occupancy models proposed by Tobler et al. (2019) to indicate differences in our notation and, more importantly, to show how parameter-expanded data augmentation (Liu & Wu, 1999) can be used to develop an efficient MCMC algorithm (Brooks et al., 2011) for fitting the MP model. In Section 2.3, we describe how analyses of simulated datasets were used to evaluate and compare the performance of the multi-species occupancy models. In Section 2.4, we describe an example wherein the MP and LV models were fit to real data collected during a camera-trap survey of mammalian species on the Tibetan Plateau.

2.1 | Multi-species occupancy models based on multivariate probit regression

2.1.1 | Modelling latent occurrences of species

Let $\boldsymbol{Z}$ denote an $n \times m$ matrix of latent species occurrences where n is the number of surveyed locations (sites) and m is the total number of distinct species observed among all locations. The value of $Z_{i,j}$ indicates whether species j was present ($Z_{i,j} = 1$) or absent ($Z_{i,j} = 0$) at site i. One or more covariates of species occurrence may be observed at each site; therefore, let $\boldsymbol{X}$ denote a $n \times p$ matrix of p regressors of species occurrence that includes a regressor for the intercept parameter (i.e. the first column of $\boldsymbol{X}$ contains only ones).

Following the approach of Chib and Greenberg (1998), we use a multivariate probit-regression model of $\boldsymbol{Z}$ to specify the effects of covariates on species occurrences in conjunction with correlations in occurrence between species. More specifically, we introduce a latent vector $\boldsymbol{u}_i = (u_{i,1}, \ldots, u_{i,m})'$ (= the ith row of $n \times m$ matrix $\boldsymbol{U}$) and make the following distributional assumptions for $\boldsymbol{u}_i$ and $Z_{i,j}$:

$$\boldsymbol{u}_i \sim \text{Normal}(\boldsymbol{\beta}'\boldsymbol{x}_i, \boldsymbol{R}) \tag{1}$$

$$Z_{i,j} \mid u_{i,j} \sim \text{Bernoulli}(I(u_{i,j} > 0)) \tag{2}$$

where $\boldsymbol{x}_i$ denotes the ith row of $\boldsymbol{X}$, $\boldsymbol{\beta}$ denotes a $p \times m$ matrix of regression parameters, $\boldsymbol{R}$ denotes a $m \times m$ symmetric matrix of correlation parameters and $I(\cdot)$ denotes the indicator function. (We use a prime superscript to denote the transpose of a matrix or vector.) The second assumption implies $Z_{i,j} = 1$ if $u_{i,j} > 0$ and $Z_{i,j} = 0$ if $u_{i,j} \leq 0$. Integrating $u_{i,j}$ from the joint density of $u_{i,j}$ and $Z_{i,j}$ reveals that marginally

$$Z_{i,j} \sim \text{Bernoulli}\left(\Phi\left(\boldsymbol{\beta}_j'\boldsymbol{x}_i\right)\right) \tag{3}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function and $\boldsymbol{\beta}_j$ is the jth column of $\boldsymbol{\beta}$. The jth column of $\boldsymbol{\beta}$ therefore contains the effects of the regressors in $\boldsymbol{X}$ on the probabilities of occurrence of species j.

Using MCMC methods to estimate $\boldsymbol{\beta}$ and $\boldsymbol{R}$ from $\boldsymbol{U}$ and $\boldsymbol{Z}$ is difficult owing to the complications of sampling correlation matrix $\boldsymbol{R}$ (Chib & Greenberg, 1998). To circumvent this problem, we adopt the approach of Zhang (2020, section 4.2) by augmenting the latent data and expanding the model's parameters to accommodate the augmented data. Specifically, we let $\boldsymbol{w}_i = \boldsymbol{D}\boldsymbol{u}_i$ where $\boldsymbol{D}$ is an $m \times m$ diagonal matrix whose nonzero elements are strictly positive (that is, $d_{j,j} > 0, \forall j$). Thus, $\boldsymbol{w}_i$ (= the ith row of $n \times m$ matrix $\boldsymbol{W}$) is a scaling of the latent data $\boldsymbol{u}_i$. Based on the distribution of $\boldsymbol{u}_i$ (Equation 1), this scaling implies $\text{E}(\boldsymbol{w}_i) = \boldsymbol{D}\boldsymbol{\beta}'\boldsymbol{x}_i$ and $\text{Cov}(\boldsymbol{w}_i) = \boldsymbol{D}\boldsymbol{R}\boldsymbol{D}$. Therefore, we can define the parameters of the augmented data as $\boldsymbol{B} = \boldsymbol{\beta}\boldsymbol{D}$ and $\Sigma = \boldsymbol{D}\boldsymbol{R}\boldsymbol{D}$ (which implies $d_{j,j} = \sqrt{\sigma_{j,j}}$, the square root of the jth diagonal element of Σ) and make the following distributional assumptions for $\boldsymbol{w}_i$ and $Z_{i,j}$:

$$\boldsymbol{w}_i \sim \text{Normal}(\boldsymbol{B}'\boldsymbol{x}_i, \Sigma) \tag{4}$$

$$Z_{i,j} \mid w_{i,j} \sim \text{Bernoulli}(I(w_{i,j} > 0)) \tag{5}$$

The assumption in Equation 5 holds because $w_{i,j} = d_{j,j}u_{i,j}$ and $d_{j,j} > 0$. Consequently, $w_{i,j} > 0$ if and only if $u_{i,j} > 0$, and $w_{i,j} \leq 0$ if and only if $u_{i,j} \leq 0$.

The distributional assumptions in Equations 4 and 5 allow us to develop an efficient MCMC algorithm for sampling the unidentified parameters $\boldsymbol{B}$ and Σ of the expanded model using the augmented data $\boldsymbol{W}$ and $\boldsymbol{Z}$. The Markov chain of $\boldsymbol{B}$ and Σ is then used to calculate the model's identified parameters as follows: $\boldsymbol{\beta} = \boldsymbol{B}\boldsymbol{D}^{-1}$ and $\boldsymbol{R} = \boldsymbol{D}^{-1}\Sigma\boldsymbol{D}^{-1}$.

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The Markov chain of these identified parameters is used to assess convergence to its stationary distribution.

2.1.2 | Modelling observed detections of species

Let $Y_{i,k,j}$ denote a binary-valued random variable whose observed value indicates detection ($Y_{i,k,j} = 1$) or non-detection ($Y_{i,k,j} = 0$) of species j during the kth survey of site i ($k = 1, \ldots, K_i$). We assume that site i is surveyed on K_i separate occasions; so let $\boldsymbol{y}_{i,j} = (Y_{i,1,j}, \ldots, Y_{i,K_i,j})'$ denote a vector of random variables whose values indicate detections or non-detections of species j during K_i repeated surveys of site i.

Covariates of detection may be measured during the repeated surveys of each site. Let $\boldsymbol{v}_{i,k} = (v_{i,k,1}, \ldots, v_{i,k,q})'$ denote a vector of q regressors of detection observed during the kth survey of site i, including a regressor for the intercept parameter (i.e. $v_{i,k,1} = 1$). If species j is present at site i, we define the conditional probability detecting this species during the kth survey of this site as follows: $p_{i,k,j} = \Phi(\boldsymbol{\alpha}_j' \boldsymbol{v}_{i,k})$, where $\boldsymbol{\alpha}_j$ is the jth column of $q \times m$ matrix $\boldsymbol{\alpha}$ of detection parameters. Each column of $\boldsymbol{\alpha}$ therefore contains the effects of the regressors on the conditional probabilities of detection of each species.

To complete the model of the observed detections, we make the following distributional assumption:

$$Y_{i,k,j} \mid Z_{i,j} \sim \text{Bernoulli}(Z_{i,j} p_{i,k,j}) \tag{6}$$

As an aside, we note that Tobler et al. (2019) specified detection parameters on the logit scale, as opposed to the probit scale. This change of scale may easily be accomplished by assuming $p_{i,k,j} = F(\boldsymbol{\alpha}_j' \boldsymbol{v}_{i,k})$, where $F(\cdot)$ denotes the standard logistic cumulative distribution function. However, we prefer to define both $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ on the probit scale by using the standard normal cumulative distribution function $\Phi(\cdot)$.

The technical details of our MCMC algorithm, which include our choice of prior distributions for the model's parameters, are described in Appendix S1. We also provide R code (R Core Team, 2025) as an implementation of this algorithm.

2.2 | Multi-species occupancy models based on latent factors

The second class of occupancy models proposed by Tobler et al. (2019) is similar to the model described in Section 2.1 except that the correlation matrix $\boldsymbol{R}$ is approximated by introducing a set of latent factors (variables) and factor loadings (parameters) into the model of $\boldsymbol{u}_i$ as follows:

$$\boldsymbol{u}_i \mid \boldsymbol{f}_i \sim \text{Normal}(\boldsymbol{\beta}' \boldsymbol{x}_i + \Lambda \boldsymbol{f}_i, \Psi) \tag{7}$$

$$\boldsymbol{f}_i \sim \text{Normal}(\boldsymbol{0}, \boldsymbol{I}_r) \tag{8}$$

where $\boldsymbol{f}_i = (f_{i,1}, \ldots, f_{i,r})'$ denotes a vector of r latent factors, Λ denotes a $m \times r$ matrix of factor loadings (the 'effects' of $\boldsymbol{f}_i$ on $\boldsymbol{u}_i$), $\boldsymbol{I}_r$ denotes the $r \times r$ identity matrix and Ψ denotes a $m \times m$ diagonal covariance matrix (for which $\psi_{j,j} > 0$, $j = 1, \ldots, m$). By integrating $\boldsymbol{f}_i$ from the joint density of $\boldsymbol{u}_i$ and $\boldsymbol{f}_i$, it is easily shown that marginally

$$\boldsymbol{u}_i \sim \text{Normal}(\boldsymbol{\beta}' \boldsymbol{x}_i, \Omega) \tag{9}$$

where $\Omega = \Lambda\Lambda' + \Psi$. As in Equation 2, we also assume

$$Z_{i,j} \mid u_{i,j} \sim \text{Bernoulli}(I(u_{i,j} > 0)) \tag{10}$$

A comparison of the distributional assumptions in Equations 1 and 2 with those in Equations 9 and 10 reveals the similarity between the occupancy models described in Sections 2.1 and 2.2. However, Ω cannot provide an approximation of $\boldsymbol{R}$ without additional constraints on Λ and Ψ.

It is well known that the parameters of a factor loadings matrix, such as Λ, must be constrained to eliminate identification problems (Rencher, 1995; Seber, 1985). Following the recommendation of Lopes and West (2004), we require Λ to be a block, lower-triangular matrix whose diagonal elements ($\lambda_{j,j}, j = 1, \ldots, r$) are strictly positive. In addition, to ensure that each diagonal element of Ω equals one, we require

$$\psi_{j,j} = 1 - \sum_{l=1}^{r} \lambda_{j,l}^2 \tag{11}$$

for $j = 1, \ldots, m$. The summation on the right hand side of Equation 11 is equivalent to the jth diagonal element of $\Lambda\Lambda'$; therefore, in matrix form this constraint is

$$\Psi = \boldsymbol{I}_m - \text{diag}(\Lambda\Lambda') \tag{12}$$

where $\boldsymbol{I}_m$ is the $m \times m$ identity matrix. The constraint in Equation 12 implies that $\Omega = \Lambda\Lambda' + \boldsymbol{I}_m - \text{diag}(\Lambda\Lambda')$ provides an approximation of $\boldsymbol{R}$. Therefore, the identifiable occurrence-related parameters of this multi-species occupancy model are $\boldsymbol{\beta}$ and the lower-triangular elements of Λ.

Tobler et al. (2019) further constrained the parameters of Λ so that $0 < \lambda_{j,j} < 1$ for diagonal elements and $|\lambda_{j,k}| < 1$ for elements below the diagonal. Tobler et al. (2019) asserted, without proof, that these constraints were 'needed for parameter identifiability reasons with binary responses'. These constraints were implemented in JAGS by restricting the prior support for each non-zero element of Λ to a compact subset of each element's theoretical support.

In conventional applications of factor analysis, the block, lower-triangular constraint on Λ places an upper bound on the number of factors wherein $r < m$ (Lopes & West, 2004). However, the additional constraint on Ψ (in Equation 12) implies that the admissible values of r must satisfy the following inequality: $m(m+1)/2 - mr + r(r-1)/2 \geq 0$. This inequality establishes that the upper bound for r is m given the additional constraint imposed on Ψ.

To complete the multi-species occupancy model based on latent factors, we account for the effects of errors in detection by using the same approach described in Section 2.1.2. We used JAGS to fit the occupancy

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model based on latent factors and to enforce the constraints on Λ recommended by Tobler et al. (2019). Our JAGS code is available online.

2.3 | Simulation study

We conducted analyses of simulated data to evaluate the performance of the two multi-species occupancy models described earlier. To distinguish between these models, we henceforth use the abbreviations MP (multivariate probit) for the model described in Section 2.1 and LF (latent factor) for the model described in Section 2.2. Although Tobler et al. (2019) used the abbreviation LV (latent variable) for the latter model, we believe that LF distinguishes these two models more accurately because both models contain latent variables but only the latter model contains latent factors.

2.3.1 | Simulating parameter values and data

To simulate a set of random species occurrences, we first selected each of m regression intercepts $(\beta_{1,1}, \ldots, \beta_{1,m})$ by drawing m values of occurrence probability ψ from a uniform(0.2, 0.8) distribution and then calculating $\beta_{1,j} = \Phi^{-1}(\psi_j)$. We also simulated each of two covariates of species occurrence by drawing random values from a uniform(−1, 1) distribution and then centring and scaling each covariate to have zero mean and unit variance. Therefore, the probabilities of species occurrence at the average value of the covariates ranged from 0.2 to 0.8. The regression coefficients for each covariate were drawn randomly from a $\text{Normal}\left(0, 0.5^2\right)$ distribution. We drew a random value of the correlation matrix $\boldsymbol{R}$ from the space of all positive-definite matrices of size $m \times m$ using the `rcorrmatrix` function of the R package `clusterGeneration` (Qiu & Joe, 2023). Finally, we drew a random value for each element of $\boldsymbol{Z}$ using the random values of $\boldsymbol{X}$, $\boldsymbol{\beta}$ and $\boldsymbol{R}$ and the underlying assumptions of the MP model (Equations 1 and 2).

To simulate a set of random detections of each species, we adopted the same approach used to simulate species occurrences. That is, we first selected each of m regression intercepts $(\alpha_{1,1}, \ldots, \alpha_{1,m})$ by drawing m values of detection probability p from a uniform(0.1, 0.7) distribution and then calculating $\alpha_{1,j} = \Phi^{-1}(p_j)$. We simulated one covariate of detection by drawing random values from a uniform(−1, 1) distribution and then centring and scaling this covariate to have zero mean and unit variance. The regression coefficients for the detection covariate were drawn randomly from a $\text{Normal}\left(0, 0.5^2\right)$ distribution. To simulate differences in survey effort among sample sites, we drew the number of surveys per site from a discrete uniform distribution of 3, 4 or 5 surveys. Finally, we drew random values for the detections of each species using the random values of $\boldsymbol{V}$, $\boldsymbol{\alpha}$ and $\boldsymbol{Z}$ as shown in Equation 6.

2.3.2 | Study design and evaluation of results

We evaluated the performance of the multi-species occupancy models by fitting the MP model and the LF model to each of 75 independent datasets that were simulated for communities of 5, 10 or 20 species surveyed at 100, 200, 400 or 800 sites. When fitting the LF model, we assigned the number of latent factors to be half the number of species. Simulation studies conducted by Tobler et al. (2019) determined that this number of latent factors was needed for the LF model to adequately approximate an unstructured correlation matrix.

To analyse the simulated data, we fit the MP model using the MCMC algorithm described in Appendix S1. We fit the LF model using the R package `rjags` (Plummer, 2025). Preliminary analysis of simulated data revealed that the MCMC algorithm used to fit the MP model was much more efficient than the algorithm produced by JAGS to fit the LF model. Consequently, we used longer Markov chains to fit the LF model than to fit the MP model. We sought to produce Markov chains whose effective sample sizes (Robert & Casella, 2004) were about the same for both models. Effective sample size (ESS) refers to the number of *independent* observations needed to obtain the same variance as that of the Markov chain, so the difference between ESS and nominal sample size (length of Markov chain) increases with the level of autocorrelation in a chain. To fit the MP model, we used 25,000 iterations of the MCMC algorithm and discarded the first 15,000 iterations (as 'burn in') to exclude initial transients in the Markov chain. The remaining 10,000 iterations of the MCMC algorithm were used without thinning to estimate summaries (means and quantiles) of the posterior distribution. To fit the LF model, we used 75,000 iterations of the JAGS-based MCMC algorithm and discarded the first 15,000 iterations as 'burn in'. The remaining 60,000 iterations were thinned by retaining every 10th iteration to produce 6000 iterations for estimating summaries of the posterior distribution. In addition to posterior summaries, we also estimated the ESS of the Markov chains retained for posterior analysis so that we could check that the simulation-based comparisons of the two models were fair.

We evaluated and compared the performance of the multi-species occupancy models in terms of their abilities to estimate the species occurrence parameters ($\boldsymbol{\beta}$), the correlations in occurrence between species ($\boldsymbol{R}$) and species detection parameters ($\boldsymbol{\alpha}$). The criteria for our evaluation included three operating characteristics: average bias, average root mean squared error (RMSE) and average coverage of 95% credible intervals. The squared error for each independent sample was calculated as the square of the Euclidean norm of $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$, a vector containing differences between estimates $\hat{\boldsymbol{\theta}}$ (posterior medians) and known values $\boldsymbol{\theta}$ of the parameters ($\boldsymbol{\beta}$, $\boldsymbol{R}$ or $\boldsymbol{\alpha}$). Mean squared error (MSE) of the estimates was then calculated using the average of the squared errors among the independent samples. RMSE was calculated using the square root of MSE. Coverage of 95% credible intervals was estimated in the usual way by determining for each independent sample whether a credible interval included the known value of a parameter. Coverage was then calculated by assuming a binomial distribution for the number intervals that successfully included the known value of the parameter among all independent samples and estimating the success probability. Preliminary analysis of simulated

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data revealed that the JAGS-based algorithm used to fit the LF model occasionally failed. Therefore, to compare the performance of LF and MP models, we used only datasets that were analysed successfully by both models.

2.4 | Case study: The Tibetan Plateau mammal community

We used the MP model and the LF model to analyse camera-trap survey data for a community of 15 mammal species detected in the Sanjiangyuan Region on the Tibetan Plateau. The data were collected as part of a long-term, community-based monitoring programme conducted between 2014 and 2018. Detections from camera traps were collected during the summer months (July to September) across 5 years at 11 monitoring sites. Each site contributed 1–3 years of data with ensured quality and covered an area of more than 3500 km^2. Within each site, camera traps were systematically arranged on a 5 × 5 km grid, with one or two cameras placed in each grid cell at locations identified by mammal signs (e.g. scat, scrapes, scent marks and pugmarks) or by distinctive topographical features (e.g. ridgelines, animal trails, trail junctions, beneath cliffs, rocky outcrops or saddles) that were likely to be visited by mammals. To reduce fine-scale spatial autocorrelation, cameras were spaced at least 1 km apart. Sampling occasions were defined as 3-day intervals to balance sample size with the explanatory power of environmental variables. We excluded 72 camera stations with fewer than five sampling occasions, as well as two species that were detected fewer than 30 times (brown bear (*Ursus arctos*) and Pallas's cat (*Otocolobus manul*)) to ensure data robustness. Owing to difficulties in distinguishing different species of pika (*Ochotona* spp.) in camera images, all pika species were treated as a single species for purposes of analysis. Therefore, the data used in our analysis included detections of 13 species at 120 camera traps. Our field work did not require permits or licences from local governments owing to our non-invasive method of sampling (camera trapping).

To account for the effects of habitat preferences, we included six environmental covariates: northness (cosine of aspect), slope, grass NDVI, forest-shrub cover, nearest distance to settlements and nearest distance to roads. Mean temperature during monitoring periods was used as a detection covariate to account for spatial and temporal variation in detection probability. For purposes of analysis, we centred and scaled each covariate to have mean zero and unit variance. Our primary goal was to investigate spatial associations among mammal species after controlling for habitat preferences and detection errors.

We fit the MP model to the data using 100,000 iterations of the MCMC algorithm. We discarded the first 50,000 iterations (as 'burn in') and retained the remaining 50,000 iterations without thinning to produce a Markov chain for estimating summaries of the posterior distribution. We fit the LF model to the data using seven latent factors (half the number of species as recommended by Tobler et al. (2019)). We used 300,000 iterations of the JAGS-based MCMC algorithm and discarded the first 50,000 iterations as 'burn in'. The remaining 250,000 iterations were thinned by retaining every 10th iteration to produce a Markov chain of 25,000 iterations for estimating summaries of the posterior distribution. These procedures yielded Markov chains for each model with similar effective sample sizes (ESS).

3 | RESULTS

3.1 | Simulation study

The results of our simulation study indicated that trends in each performance measure (bias, RMSE and coverage) of parameter estimates with increasing sample size (number of locations surveyed) were similar for both MP and LF models. For example, average biases in estimates of occurrence parameters, correlation parameters and detection parameters did not differ significantly from zero (Figure 1), and estimates of average RMSE declined progressively with increasing sample size (Figure 2). For occurrence parameters and detection parameters, estimates of average coverage of 95% credible intervals increased with sample size and approached or reached the nominal level of 0.95 (Figure 3). In contrast, estimates of average coverage for correlation parameters progressively declined below the nominal level as sample size was increased. The magnitude of these declines was greater for correlations estimated with the MP model than for those estimated with the LF model.

To explore potential reasons for the declines in coverage of correlation parameters, we plotted the bias in each estimated correlation against the known value of this parameter. We examined biases based on fitting the two occupancy models (MP or LF) to observable data and fitting a multivariate probit regression model (MvP) to latent species occurrences. Figure 4 illustrates an example of these plots for the case of $m = 10$ species and $n = 400$ survey locations. These plots reveal a systematic bias in correlations estimated with the MP or LF models, wherein strongly negative correlations were overestimated and strongly positive correlations were underestimated. In other words, estimates of strongly negative correlations were not as low as they should be, and estimates of strongly positive correlations were not as high as they should be. Evidently, these two opposing sources of bias had been masked when calculating the *average* bias among all correlation parameters (as in Figure 1) because the opposing sources of bias effectively cancelled one another. Although systematic bias in correlation estimates was present in both occupancy models, the effects of this bias on coverage were more evident in the MP model than in the LF model (Figure 3), primarily because correlations estimated with the MP model were generally more precise than those estimated with the LF model (Figure 4). In contrast to the occupancy models, correlations estimated with the MvP model contained no systematic bias (Figure 4), and average coverages of 95% credible intervals estimated for correlation parameters and for occurrence parameters increased with sample size and reached the nominal level of 0.95 (Figure S1).

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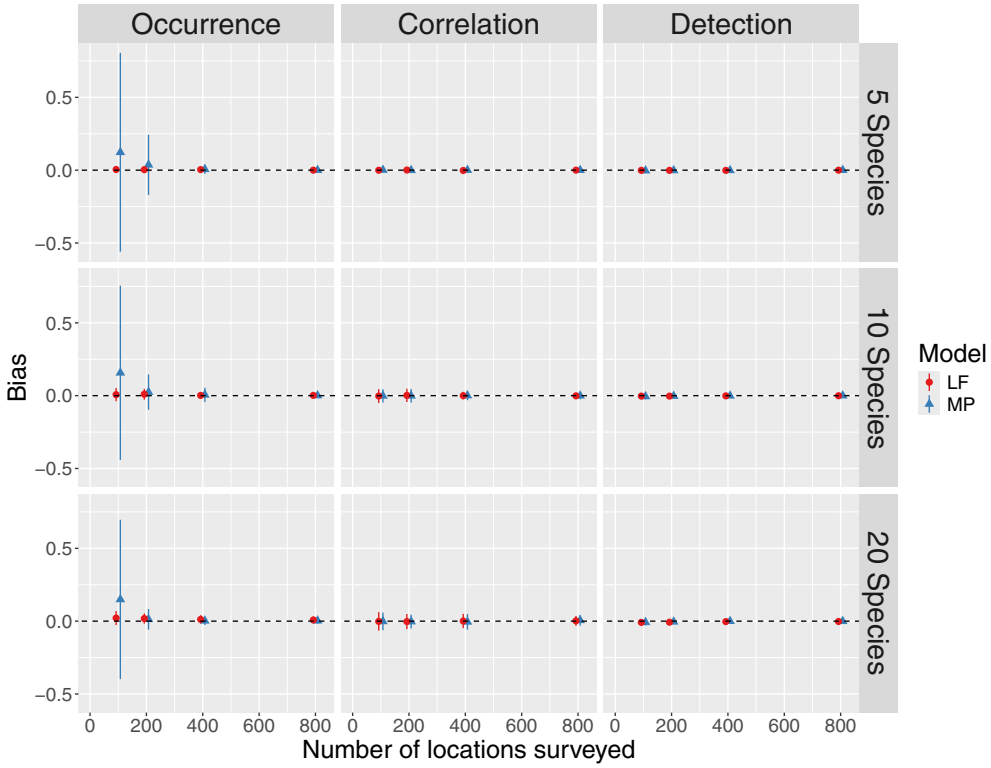


FIGURE 1 Average bias (mean and 95% confidence interval) of estimates of species occurrence (β), correlation (R) and detection (α) parameters obtained by fitting MP models (blue triangles) or LF models (red circles) to simulated datasets.

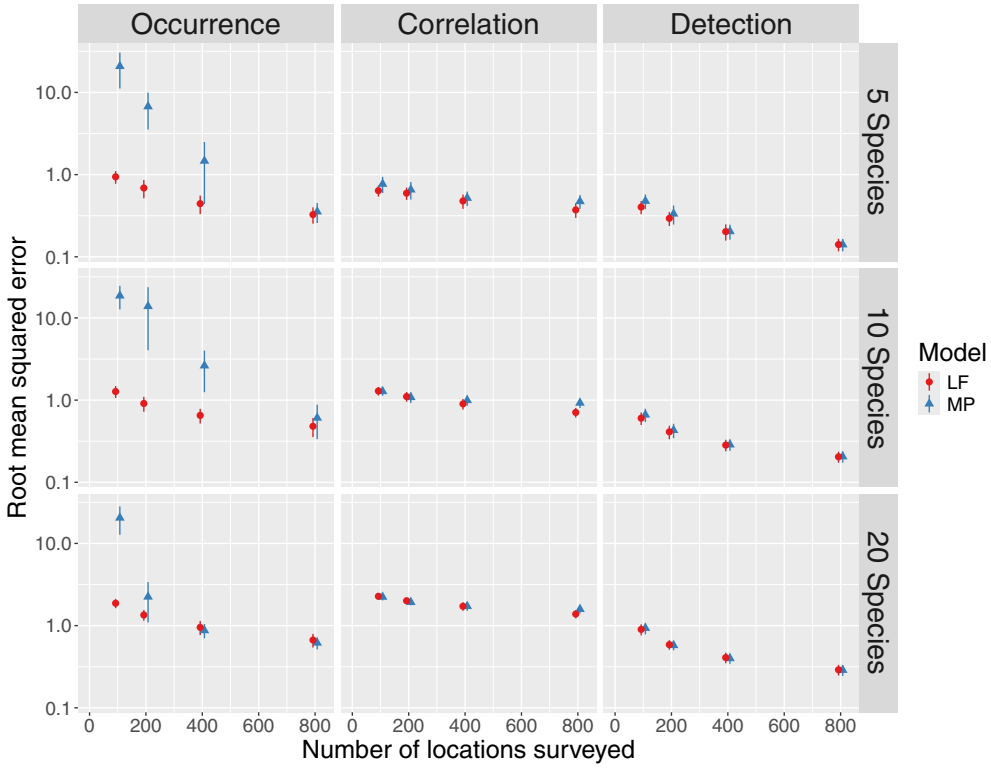


FIGURE 2 Average RMSE (mean and 95% confidence interval) of estimates of species occurrence (β), correlation (R) and detection (α) parameters obtained by fitting MP models (blue triangles) or LF models (red circles) to simulated datasets.

Our simulation study also revealed differences in the Markov chains produced by the MP model's MCMC algorithm (Appendix S1) and the LF model's JAGS-based MCMC algorithm. As mentioned earlier, levels of autocorrelation were consistently higher in Markov chains computed with the LF model's algorithm than in those computed with the MP model's algorithm. We had deliberately increased

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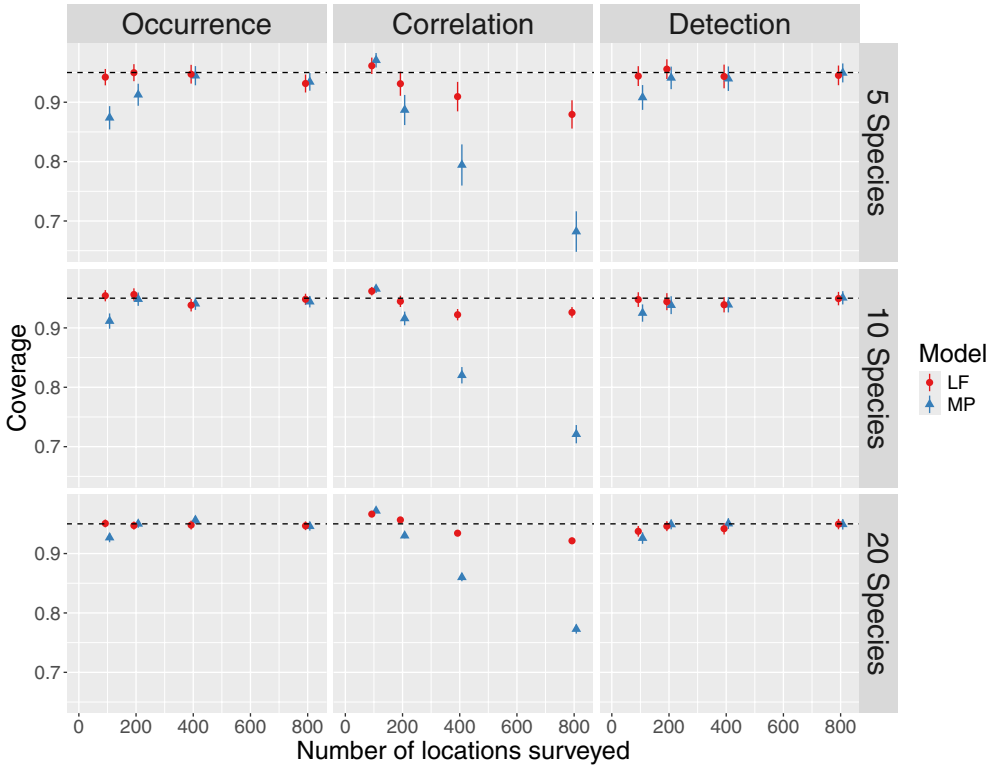


FIGURE 3 Average coverage (mean and 95% confidence interval) of 95% credible intervals for species occurrence ($\boldsymbol{\beta}$), correlation ($\boldsymbol{R}$) and detection ($\boldsymbol{\alpha}$) parameters obtained by fitting MP models (blue triangles) or LF models (red circles) to simulated datasets. Horizontal dashed line indicates the nominal level of coverage.

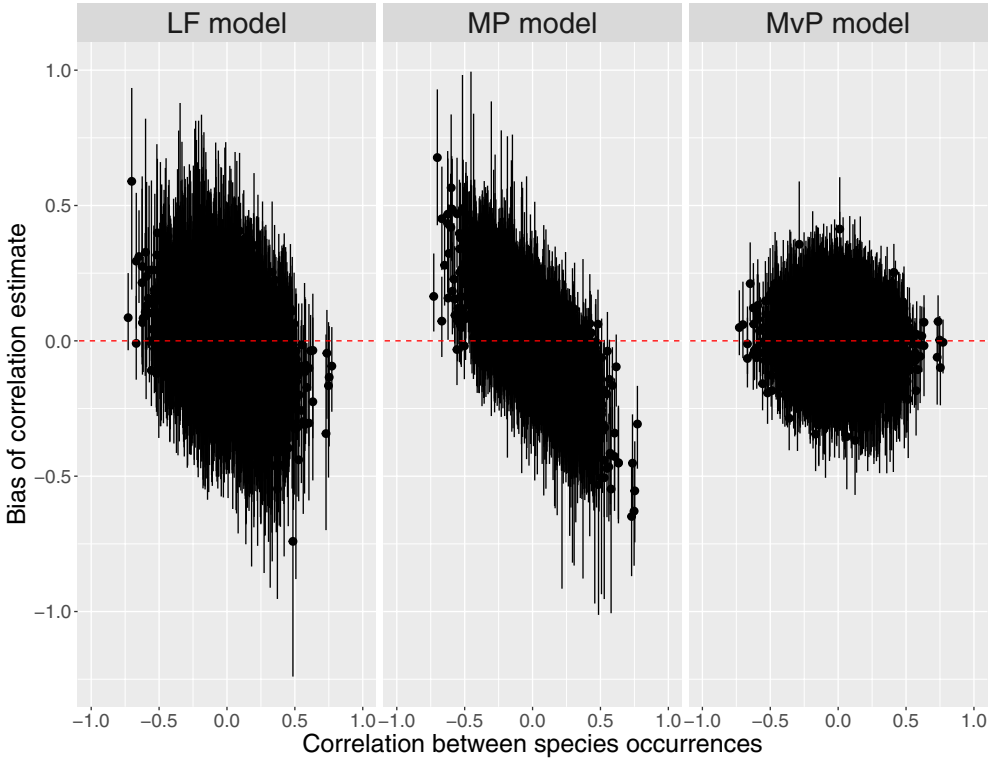


FIGURE 4 Bias (median and 95% credible interval) of correlation parameters estimated by fitting occupancy models (MP or LF) to simulated detection data or by fitting a multivariate probit regression model (MvP) to simulated species occurrences. The data and occurrences were obtained in simulations of $m = 10$ species and $n = 400$ survey locations.

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the length and thinning rate of Markov chains used to fit the LF model to achieve comparable levels of effective sample size for both LF and MP models. Despite these attempts to achieve comparable results, the LF model's MCMC algorithm failed in 13% of the simulated datasets (95% confidence interval = 11%–15%). Failure rate did not differ significantly with number of species or number of locations surveyed ($p = 0.46$). In contrast, the MP model's MCMC algorithm was successful in analysing all of the simulated data sets.

3.2 | Case study: The Tibetan Plateau mammal community

Fitting the MP model to the camera-trap survey data revealed 12 significant correlations in occurrence among the 78 pairs of 13 species (Figure 5). The LF model identified nine significant correlations (Figure 5), eight of which were found using the MP model. The additional significant correlation (estimated to be 0.33 between stone marten [*Martes foina*] and Altai weasel [*Mustela altaica*] occurrences) was similar in magnitude to that estimated by the MP model (0.25) but was not statistically significant in the latter model. Estimates of posterior summary statistics for correlations between all 78 pairs of species are reported in Table S1 (MP model) and Table S2 (LF model).

Among the significant correlations, occurrences of grey wolves (*Canis lupus*) were positively associated with those of two mesopredators (stone martens and Tibetan foxes [*Vulpes ferrilata*]) according to the results of both models. Occurrences of grey wolves also were positively associated with those of two large herbivores (alpine musk deer [*Moschus chrysogaster*] and white-lipped deer [*Przewalskium albirostris*]) according to the MP model's results. Pika spp. (*Ochotona* spp.) were positively associated with three mesopredators (Altai weasels, stone martens and red foxes [*Vulpes vulpes*]) according to the results from both models. The MP model's results also revealed an unexpected positive association between occurrences of pika spp. and blue sheep (*Pseudois nayaur*). Additionally, occurrences of white-lipped deer were positively associated with those of alpine musk deer according to the MP model's results, and unexpectedly, with those of red foxes according to the results of both models. Finally, occurrences of Tibetan foxes exhibited positive associations with occurrences of stone martens and domestic yaks according to the results of both models.

4 | DISCUSSION

We devised an efficient MCMC algorithm for fitting the MP model using parameter-expanded data augmentation. This algorithm allowed us to conduct a fair comparison of the two multi-species occupancy models with correlated species occurrences proposed by Tobler et al. (2019). In contrast to their findings, the results of our simulation study revealed that model-specific estimators of species occurrence, correlation and detection parameters had similar performance in terms of average bias, average RMSE and average coverage of 95% credible intervals regardless of whether the MP model or the LF model was used to analyse the data. These results were actually expected because we had fit the LF model using a relatively high number of latent factors (one half of the number of species), as recommended by Tobler et al. (2019). Using many latent factors therefore had improved the accuracy of the LF model's approximation of the species correlation matrix.

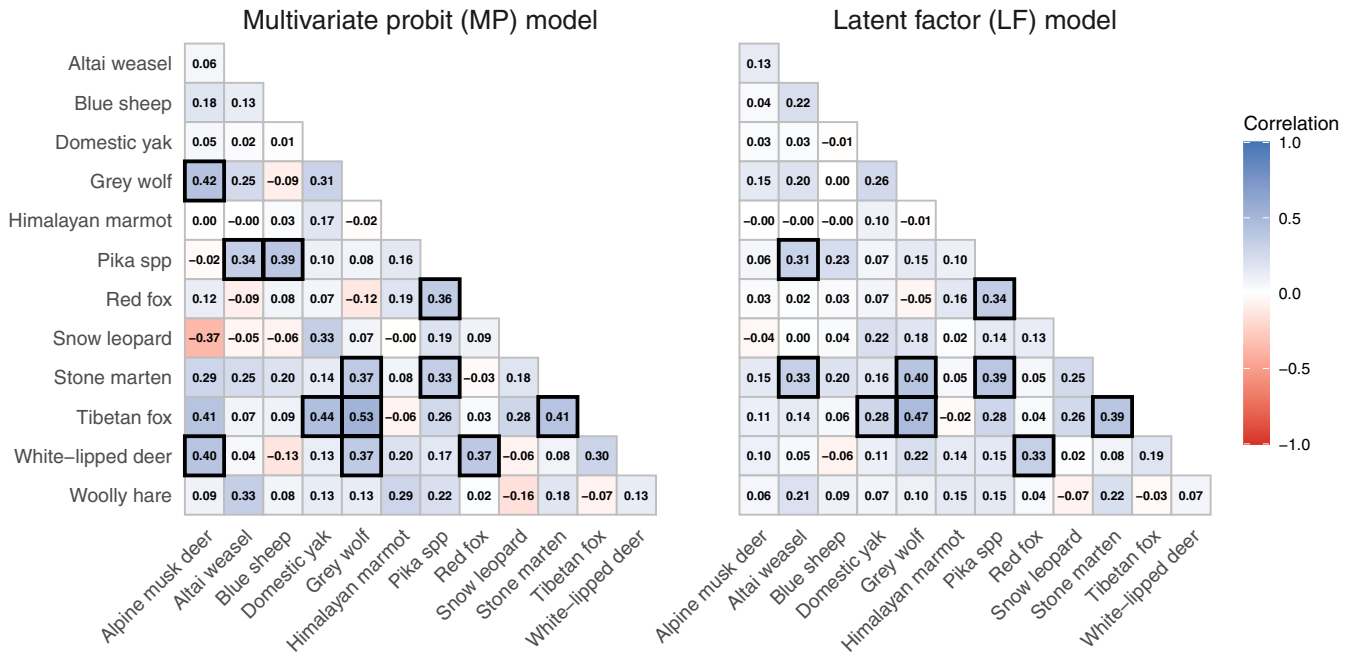


FIGURE 5 Estimated correlations in occurrence between 13 mammal species in the Sanjiangyuan Region on the Tibetan Plateau. The black border indicates correlations whose 90% credible interval did not include zero.

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Our simulation study also revealed the presence of systematic bias in estimates of the correlation parameters, wherein strongly negative correlations were overestimated and strongly positive correlations were underestimated. This bias was present in the estimates of both occupancy models (MP and LF). To identify the source of bias, we fit a multivariate probit regression (MvP) model to the species occurrences simulated in our study. The full conditional distributions used to estimate the parameters of the MvP model ($\boldsymbol{\beta}$ and $\boldsymbol{R}$) are *identical* to those used to estimate these parameters in the MP occupancy model. Therefore, the only difference between these two models is that when fitting the MvP model, the values of species occurrence (in matrix $\boldsymbol{Z}$) are treated as fixed observations; whereas, when fitting the MP model, elements of $\boldsymbol{Z}$ must be estimated at locations where a species is not detected. Our simulation results revealed that correlations estimated with the MvP model displayed no signs of systematic bias. Furthermore, average coverages of the correlation parameters reached the nominal level (0.95) as sample size was increased. Therefore, systematic biases in correlations estimated with MP or LF occupancy models must have been produced by the failure of these models to estimate species occurrences accurately. This shortcoming of the MP and LF occupancy models is a new finding that was not identified the simulation studies of Tobler et al. (2019).

In our case study of the Tibetan Plateau mammal community, estimates of correlations in occurrence between species were largely consistent between the MP and LF models, although the MP model's fit often produced stronger associations between species. These differences may have been attributed to the fact that the MP model's parameters include the actual correlations between species, whereas the LF model's parameters approximate these correlations using a factor-analytic approximation of the correlation matrix. Many of the significant species associations identified with the MP model align well with ecological knowledge. For example, both alpine musk deer and white-lipped deer have been reported as prey for grey wolves (Chetri et al., 2017; Lu et al., 2023; Mahmood et al., 2021). Similarly, pika spp. are recognized as primary prey for Altai weasels, stone martens and red foxes (Bakaloudis et al., 2012; Irshad et al., 2021; Roy et al., 2019; Tsukada et al., 2014), the three meso-predators in our study. Tibetan foxes also have been frequently observed using domestic yaks as shelter while hunting pikas, often appearing within yak herds (BBC, 2024).

However, some of the positive associations between species identified in our analysis may reflect similar habitat preferences rather than direct interactions between species. For example, alpine musk deer and white-lipped deer both inhabit alpine shrubland-meadows or open woodlands. Likewise, the co-occurrence of Tibetan foxes and stone martens may be attributed to shared prey species, such as pikas and rodents, and the co-occurrences between these meso-predators and grey wolves could reflect shared resources, such as food or available burrows (Wang & Hou, 2021). As stated earlier, caution is warranted when interpreting correlations in occurrence between species because the source of these correlations can include interactions between species or the effects of unmeasured environmental covariates. For example, in our case study, fine-scale environmental variables, such as prey abundance and burrow density, were neither measured nor included in the analysis, and the effects of these variables may have contributed to some of the positive associations between species.

Our analyses of real and simulated data suggest that either the MP model or the LF model can be used to estimate species occurrences, the effects of spatially referenced covariates of species occurrence and the correlations in occurrence between species; however, some correlation estimates may include substantial bias if the species detection probabilities are low. An advantage of the LF model is that the number of parameters needed to approximate the correlation matrix $\boldsymbol{R}$ will be much lower than $m(m-1)/2$ if the number of latent factors r is much lower than m. The reason for this advantage is that the number parameters in Λ, the $m \times r$ block, lower-triangular matrix used to compute an approximation of $\boldsymbol{R}$, does not increase quadratically with m when $r \ll m$. However, selecting the LF model in favour of the MP model also comes with disadvantages. First, an analyst must decide whether the number of latent factors used to fit the LF model is sufficiently high to provide a satisfactory approximation of the correlation matrix $\boldsymbol{R}$. This decision typically involves selecting several values of r, using each value to fit the LF model and estimating $\boldsymbol{R}$ from each fitted LF model. Estimates of $\boldsymbol{R}$ should become similar as the number of latent factors increases; otherwise, the number of latent factors must be increased further and the process repeated until a satisfactory approximation of $\boldsymbol{R}$ is obtained. Tobler et al. (2019) proposed an alternative diagnostic (residual sum of squares) for deciding whether the number of latent factors is adequate; however, the LF model still must be fitted several times (one fit for each candidate value of r) to determine whether any of the candidates is satisfactory. Another disadvantage of selecting the LF model is associated with the JAGS-based MCMC algorithm used to fit this model. In our simulation study, this algorithm produced Markov chains that were highly autocorrelated and required many more iterations and thinning to achieve effective sample sizes that were comparable to those of unthinned Markov chains produced by the MP model's MCMC algorithm. In addition, the LF model's MCMC algorithm was not always successful, failing in 13% of attempts to fit the model. Other researchers have reported difficulties with the LF model's MCMC algorithm while attempting to fit the LF model to real data. For example, Kéry & Royle (2021, section 8.4.2) were only able to fit the LF model with $r = 2$ factors to a subset of the data analysed by Tobler et al. (2019) (surveys of 30 species at 1200 locations). Kéry and Royle (2021) noted that Markov chains for the factor-loadings parameters (elements of Λ) were 'mixing very badly' and that attempts to fit the LF model 'lasted several days'. Given that the Markov chains for factor loadings parameters were evidently highly autocorrelated and that the LF model contained only two latent factors, the accuracy of the estimated correlations in occurrence between species (Kéry & Royle, 2021, fig. 8.16) is questionable. In summary, the LF model may include fewer estimable parameters than the MP model (especially in communities of many species), but the JAGS-based MCMC algorithm used to fit

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the LF model is not without its problems. Furthermore, analysts also must determine whether the number of latent factors is sufficiently high to provide an adequate approximation of the correlations in occurrence among species.

The MP and LF models described in Sections 2.1 and 2.2 differ slightly from those proposed by Tobler et al. (2019), who added the assumption of exchangeability for species-specific occurrence and detection parameters and noted that these parameters were 'modeled hierarchically among species'. Exchangeability among species is sensible to assume in a community of ecologically similar species. In fact, exchangeability is necessary if estimates of biodiversity are needed (Dorazio et al., 2006, 2011; Dorazio & Royle, 2005; Gelfand et al., 2005) because this assumption permits the contributions of unobserved species to be included in the estimates. However, exchangeability is not an essential component of the models proposed by Tobler et al. (2019). We purposely excluded this assumption from the MP and LF models described in Sections 2.1 and 2.2 because doing so would restrict application of these models to communities of ecologically similar species. It is entirely appropriate to fit these models to data collected in presence–absence surveys of communities of ecologically, and perhaps taxonomically, dissimilar species. Nowadays, such surveys are commonplace owing to the widespread use of motion-activated cameras for detection of species, as in our case study. Therefore, the MP and LF models described in Sections 2.1 and 2.2 can be used to infer correlations in occurrence that may be associated with interactions between species, such as competition for resources, interference competition or predator–prey interactions. However, as described earlier, these correlations are simply measures of co-occurrence between species that may be attributed to interactions between species or to the effects of unobserved environmental covariates on species occurrences.

Our evaluation did not include the class of multi-species occupancy models developed recently by Doser et al. (2023). These models are quite versatile because they can be used to estimate several sources of variation in species occurrences, including effects of environmental covariates, effects of unobserved sources of spatial dependence and effects of correlations between species. However, these models also use an arbitrarily specified number of latent factors to approximate correlations in occurrence among species, so an analyst must consider models with different numbers of latent factors, as described earlier for the LF model. Doser et al. (2023) acknowledged this aspect of their models and noted that 'Determining the optimal number of factors for a given data set is not straightforward and will vary depending on the characteristics of the specific community of species'. An important distinction between the occupancy models of Doser et al. (2023) and those of Tobler et al. (2019) is that the former models assume the presence of residual variation in species occurrences that stems from overdispersion (i.e. extra-Bernoulli variation). In contrast, by adopting the multivariate probit regression model of Chib and Greenberg (1998), the models of Tobler et al. (2019) specify correlations in occurrence between species without assuming overdispersion (residual variation) in the marginal distribution of each species' occurrence. Therefore, marginal probabilities of species occurrence based on these models depend only on the occurrence parameters (β). In contrast, in the models of Doser et al. (2023) marginal probabilities of species occurrence depend on the occurrence parameters *and* on the residual variance parameters that quantify the levels of overdispersion. Furthermore, this dependence extends to the marginal variances and covariances of species occurrence. Doser et al. (2023) conducted simulation studies to compare estimates of species occurrence based on coverage of occurrence-related parameters, but coverage of correlation parameters was not included in these studies. Therefore, it is unknown whether systematic bias exists in estimates of correlations obtained by fitting the models of Doser et al. (2023).

Although the models of Tobler et al. (2019) and Doser et al. (2023) are related, they also possess inherent differences. Both are potentially useful; so how should an analyst select either model in favour of the other? The answer largely depends on the data and on the levels of overdispersion in species occurrences. If most of the spatial variation in species occurrences can be accounted for by informative covariates of occurrence, the models of Tobler et al. (2019) are likely to be useful and satisfactory. On the other hand, if these covariates fail to account for the spatial variation in species occurrences, the models of Doser et al. (2023) are likely to be needed. As is often true, familiarity with the data is paramount in selecting a model for the estimation or prediction of species occurrences.

Our evaluation also did not include multi-species occupancy models that specify co-occurrences between species that depend on the occupancy state (presence or absence) of other species. Such models were first developed by MacKenzie et al. (2004) and then reparameterized for specific patterns of co-occurrence wherein some species are assumed to be dominant (their occurrence is not influenced by the occupancy state of other species) while other species are assumed to be subordinate (their occurrence depends on the occupancy state of dominant species) (Richmond et al., 2010; Waddle et al., 2010). These patterns of co-occurrence were intended to specify the effects of asymmetric interactions between species, such as interactions between predators and prey. Following these early models of co-occurrence, Rota et al. (2016) exploited the newly developed multivariate Bernoulli distribution (Dai et al., 2013) and developed a multi-species occupancy model that allowed the effects of symmetric or asymmetric interactions between species to be specified. In principle, this multi-species occupancy model can be used to specify the effects of interactions among any number of species; however, in practice, attempts to fit models of more than about 10 species are prone to failure (Kéry & Royle, 2021, section 8.2). This deficiency obviously limits applications of this potentially useful class of occupancy models.

We conclude by noting that ecologists are now armed with several options for analysing presence–absence surveys of entire communities, including those with species whose occurrences may be correlated. The multi-species occupancy models of Rota et al. (2016), Tobler et al. (2019) and Doser et al. (2023) each possess strengths and weaknesses, but taken together these models provide a variety of opportunities for analysing community-level, presence–absence

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surveys while accounting for the effects of false-negative errors in detection.

AUTHOR CONTRIBUTIONS

Robert M. Dorazio provided conceptualization, developed the statistical methods and wrote the code to implement these methods. Peiyun Li and Lingyun Xiao collected and organized the data. Robert M. Dorazio, Peiyun Li and Shannon M. Kachel analysed the data. Robert M. Dorazio led the writing of the manuscript. All authors contributed critically to drafts of the manuscript and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

Data and code are available via https://doi.org/10.5281/zenodo.17178936 (Dorazio et al., 2025) or via https://github.com/RobertDorazio/MultispeciesOccupancy.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. MCMC algorithm used to fit the multivariate probit (MP) occupancy model.

Figure S1. Average coverage of credible intervals for species occurrence and correlation parameters obtained by fitting a multivariate probit regression (MvP) model to simulated species occurrences.

Table S1. Estimates of posterior summary statistics for correlations in occurrence between 13 mammalian species on the Tibetan Plateau. Estimates were computed by fitting the MP model.

Table S2. Estimates of posterior summary statistics for correlations in occurrence between 13 mammalian species on the Tibetan Plateau. Estimates were computed by fitting the LF model.

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